One Step Function Explains Why Galaxies Die

Gas supply versus gas depletion timescales set a universal threshold for quenching and rejuvenation.

arXiv:2610.20505 — Step-like quenching from gas mass conservation in 350,000 galaxies

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Background

Galaxies stop forming stars through many different mechanisms, yet no simple framework unites them. Gas mass conservation implies quenching follows a step-function response to supply versus depletion timescales.

> **A unified, physically motivated link between black hole feedback, halo growth, and observed quenching patterns is still missing.**

What this paper is selling

The authors infer gas-supply timescales directly from quenched fractions, finding a characteristic 1 Gyr. Black holes drive inside-out quenching; halos drive outside-in, with intermediate-density galaxies sitting at a critical tipping point.

Key results to remember

- Characteristic gas-supply timescale is 1 Gyr across 347,182 low-redshift galaxies.
- Black hole growth lengthens and narrows supply timescales; halo mass lengthens and broadens them.
- Intermediate surface-density galaxies are critical, most sensitive to quenching and rejuvenation.

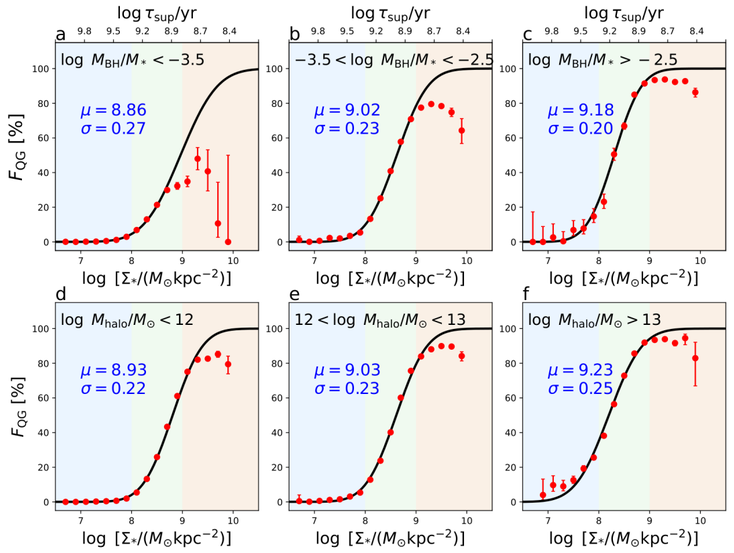


Figure 2 | Black hole and halo mass shift supply timescales in opposite ways, revealing two distinct regulators of quenching.

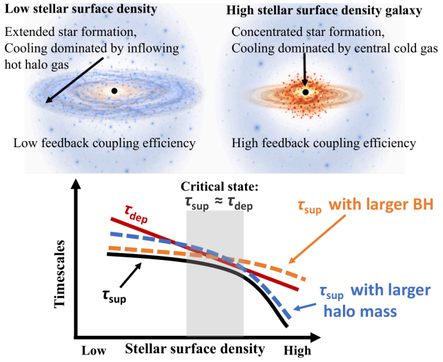


Figure 4 | A physical picture: halo cooling governs extended star formation, while black hole feedback targets centrally concentrated gas.